

8.03 Groups			
8.03a	Binary operations	a) Be able to work with binary operations and their properties when defined on given sets. <i>Includes knowing and understanding the terms “associativity” and “commutativity”.</i>	
8.03b		b) Be able to construct Cayley tables for given finite sets under the action of a given binary operation. <i>Multiplicative notation and/or terminology will generally be used, when appropriate.</i>	

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
8.03c	Definition of a group	c) Recall and be able to use the definition of a group, for example to show that a given structure is, or is not, a group. <i>e.g. Questions may be set on groups of integers modulo n (for $n \geq 2$), functions, matrices, transformations, the symmetries of given geometrical shapes and complex numbers.</i> <i>Groups may be referred to in either of the forms:</i> 1. <i>by the given set and associated binary operation $(G, \circ)$,</i> 2. <i>as “G”, where the operation is understood, or</i> 3. <i>as “the set G with the operation $\circ$”.</i> <i>To include knowing the meaning of the terms “identity” and “closed”, and that in an abelian group the operation is commutative.</i>	
8.03d		d) Recognise and be able to use the Latin square property for group tables.	
8.03e	Orders of elements and groups	e) Recall the meaning of the term “order”, as applied both to groups and to elements of a group, and be able to determine the orders of elements in a given group. <i>Includes knowing and being able to use the fact that the order of an element is a factor of the order of the group.</i>	
8.03f	Subgroups	f) Understand and be able to use the definition of a subgroup, find subgroups and show that given subsets are, or are not, proper subgroups.	
8.03g	Cyclic groups	g) Recall the meaning of the term “cyclic” as applied to groups.	

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8.03h	Generators	h) Understand that a cyclic group is generated by “powers” of a single element (generator), that there may be more than one such element within a group, and that other (non-cyclic) groups may be generated by two or more elements along with their “powers” and “products”.	
8.03i 8.03j	Properties of groups	i) Be familiar with the structure of finite groups up to, and including, order seven, and be able to apply this knowledge in solving problems.	j) Be able to work with groups of higher finite order, or of infinite order. <i>No particular prior knowledge of specific groups, or their structures, will be expected.</i>
8.03k	Lagrange’s theorem		k) Recall and be able to apply Lagrange’s theorem concerning the order of a subgroup of a finite group. <i>The proof of the theorem is not required, but a clear statement of the result may be expected.</i>
8.03l	Isomorphism		l) Be able to determine whether two given groups are, or are not, isomorphic using informal methods. <i>e.g. By noting disparities between the orders of elements.</i>
8.03m	Abstract groups		m) Be able to work with groups defined by their algebraic properties. <i>Includes using algebraic methods to establish properties in abstract groups.</i> <i>e.g. To show that any group in which every element is self-inverse is abelian, to establish an identity or complete a Cayley table.</i>